Chayan Roychoudhury

Contact Information

Modeling and Analysis of Atmospheric Composition Laboratory

Education

PhD, Atmospheric Sciences — **The University of Arizona** — 2021 – 2025
Supervisor: Dr. Avelino F. Arellano Jr.

MSc, Atmospheric Sciences — **University of Calcutta** — 2017 – 2019
Thesis: Simulation of Hygroscopic Factors on Polar Aerosols over East Antarctica

BSc, Physics — **University of Calcutta** — 2014 – 2017

Work Experience

Postdoctoral Research Associate — Oct 2025 - *Present*
Graduate Research Assistant — Jan 2021 - *Sep 2025*
Department of Hydrology and Atmospheric Sciences, The University of Arizona
Supervisor : Dr. Avelino F. Arellano Jr.

Guest Research Worker — Aug 2019 - Jul 2020
Bose Institute, Environmental Sciences Section
Supervisor : Dr. Sanat Kumar Das

Teaching Experience

ATMO 430 - Computational Methods in Atmospheric Sciences (Spring 2023)
ATMO 545 - Introduction to Data Assimilation (Spring 2024)
ATMO 469A/569A - Air Pollution I : Gases (Fall 2024, Fall 2025)
ATMO 469B/569B - Air Pollution II : Aerosols (Spring 2026)

Research Experience

Publications

1. JH Richter, E Joseph, MC Arcodia, J Berner, JL Demuth, P Falloon, GS Romine, JT Cohen, J Gonzalez-Cruz, M El Gharamti, B Hoppe, S Kumar, A Mariotti, D Mishra, K Pegion, Z Pu, KA Quagraine, KT Quagraine, *C Roychoudhury*, J Ryan, Z Stone, D Das, B Gaubert, S Kapnick, C Zarzycki. 2026. **Earth System Predictability Across Time Scales for a Resilient Society: A Research Community Perspective**. *Bulletin of the American Meteorological Society*.
   JM McKinnon, *C Roychoudhury*, and AF Arellano Jr: **Spatio-temporal pattern analysis of MOPITT total column CO using varimax rotation and singular spectrum analysis**. *Atmospheric Measurement Techniques*. *(Accepted)*.
2. *C Roychoudhury*, C He, R Kumar, AF Arellano Jr. 2025. **Diagnosing Aerosol-Meteorological Interactions on Snow within the Earth System: A Proof-of-Concept Study over High Mountain Asia**. *Earth System Dynamics*.
3. *C Roychoudhury*, C He, R Kumar, JM McKinnon, and AF Arellano. **On the relevance of aerosols to snow cover variability over High Mountain Asia.** 2022. *Geophysical Research Letters, 49*.
4. R Kumar, C He, *C Roychoudhury*, W Cheng, N Mizukami, and AF Arellano. **High Mountain Asia 12 km Modeled Estimates of Aerosol Transport, Chemistry, and Deposition Reanalysis, 2003-2019. (HMA2_MATCHA, Version 1)**. 2024. *[Data Set]*. Boulder, Colorado USA. *NASA National Snow and Ice Data Center Distributed Active Archive Center*.
5. K Mottungan, *C Roychoudhury*, V Brocchi, B Gaubert, W Tang, MA Mirrezaei, JM McKinnon, Y Guo, DWT Griffith, DG Feist, I Morino, MK Sha, MK Dubey, M De Maziere, NM Deutscher, PO Wennberg, R Sussmann, R Kivi, TY Goo, VA Velazco, W Wang, and AF Arellano Jr. **Local and regional enhancements of CH4, CO, and CO2 inferred from TCCON column measurements**. 2024. *Atmospheric Measurement Techniques, 17*, 5861–5885.
6. M Greenslade, Y Guo, G Betito, MA Mirrezaei, *C Roychoudhury*, AF Arellano, A Sorooshian. **On ozone's weekly cycle for different seasons in Arizona.** 2024. *Atmospheric Environment*.
7. MA Mirrezaei, AF Arellano, Y Guo, *C Roychoudhury*, A Sorooshian. **Ozone Production over Arid Regions: Insights into Meteorological and Chemical Drivers**. 2024. *Environmental Research Communications*.
8. Y Guo, *C Roychoudhury*, MA Mirrezaei, R Kumar, A Sorooshian, and AF Arellano: **Investigating Ground-Level Ozone Pollution in Semi-Arid and Arid Regions of Arizona Using WRF-Chem v4.4 Modeling**. 2024. *Geoscientific Model Development*, 17, 4331–4353.

9. A Sorooshian, AF Arellano Jr, M Fraser, P Herckes, G Betito, E Betterton, R Braun, Y Guo, MA Mirrezaei, *C Roychoudhury*. **Ozone in the Desert Southwest of the United States: A Synthesis of Past Work and Steps Ahead.** 2024. *ACS ES&T Air, 1(2), 62-79*.
10. D Das, S Chiao, *C Roychoudhury*, F Khan, S Chaudhuri, S Mukherjee. **Tropical Cyclone Energy Variability in North Indian Ocean: Insights from ENSO.** 2023. *Climate, 11, 232*.

In progress

1. N Mizukami, S Minallah, C He, *C Roychoudhury*, W Cheng, R Kumar. **Hydrologic implications of aerosol deposition on snow in High Mountain Asia river basins**. *Hydrology and Earth System Sciences*. (*In review*).
2. JM McKinnon, *C Roychoudhury*, MA Mirrezaei, B Gaubert, AF Arellano. **Dominant Time Scales of CO Variability from Satellites and Models Using Power Spectral Analysis**. *JGR Atmospheres*. (*In review*).
3. *C Roychoudhury*, W Cheng, N Mizukami, C He, AF Arellano and R Kumar. **MATCHA, Model for Atmospheric Transport and Chemistry in Asia, a novel regional climate-chemical reanalysis. Part 1: System description and initial evaluation**. *Earth System Science Data*. (*In review*).
4. *C Roychoudhury*, C He, R Kumar, and AF Arellano. **Revisiting Black Carbon Emission Estimates in the Third Pole: A Hierarchical Bayesian Synthesis Inversion Perspective.** Environmental Research Communications. *(In review)*.
5. H Bai, *C Roychoudhury*, C Strong. **Variability of summertime North American Dipole from large ensemble climate simulations.** (*In preparation*).
6. SK Das, *C Roychoudhury*, and A Taori. **Virga observed over East Antarctica: An alarming indication of global warming**. (*In preparation*).

Conference Presentations

1. *C. Roychoudhury*, MA Mirrezaei, K Pan, Y Guo, A Sorooshian, E Robinson, MP Fraser, P Herckes, and AF Arellano Jr. **Towards Constraining VOC Emissions over Phoenix during the 2025 GLOR Campaign with Ensemble Models**. AMS 2026, Houston, TX.
2. ARW Arellano and *C. Roychoudhury*. **Investigating Air Quality in Tucson Using Decades of Surface Observations**. AMS 2026, Houston, TX.
3. M Robertson, *C Roychoudhury*, MA Mirrezaei, A Sorooshian, and AF Arellano Jr. **Diurnal and Seasonal Variability of Ozone Precursors Across Arizona's Sun Corridor**. AMS 2026, Houston, TX.
4. K Pan, *C Roychoudhury*, BG Beran, F Chouza, T Leblanc, MP Fraser, P Herckes, A Sorooshian, and AF Arellano Jr. **A Stratospheric Intrusion Event Contributing to Elevated Surface Ozone Over the Phoenix Metropolitan Region in June 2025**. AMS 2026, Houston, TX.
5. *C Roychoudhury*, AF Arellano. **Diagnosing Interactions at the Earth System Interface: Case Studies from Snowmelt and Chemical Feedbacks.** Poster NG43A-2301 presented at AGU Fall Meeting 2024, Washington DC.
6. *C Roychoudhury*, JM McKinnon, MA Mirrezaei, Y Guo, AF Arellano. **Source attribution of wildfires in Arizona: An exploration of the June 2021 wildfire season**. Poster GC23F-0335 presented at AGU Fall Meeting 2024, Washington DC.
7. *C Roychoudhury*, MA Mirrezaei, Y Guo, AR Arellano, G Betito, A Sorooshian, AF Arellano. **Leveraging atmospheric chemistry observations in Arizona: Insights into the regional transport of ozone and aerosols.** Poster presented at AERONET Science and Application Exchange, College Park, MD, September 17-19, 2024.
8. *C Roychoudhury*, AF Arellano, W Cheng, N Mizukami, J McKinnon, C He, R Kumar. **Is traditional Bayesian inversion sufficient to constrain black carbon abundance in High Mountain Asia?** Poster C84 presented at the iCACGP-IGAC 2024 Conference in Kuala Lumpur, Malaysia, 9th-13th September, 2024.
9. *C Roychoudhury*, W Cheng, C He, R Kumar, JM McKinnon, AF Arellano. **How uncertain are BC emissions in High Mountain Asia? An inverse modeling approach.** Poster GC21K-1070 at AGU Fall Meeting 2023, San Francisco, CA.
10. *C Roychoudhury*, W Cheng, C He, R Kumar, AF Arellano. **MATCHA, Model for Atmospheric Transport and Chemistry in Asia: A novel regional climate-chemical reanalysis.** Poster C51C-0957 at AGU Fall Meeting 2023, San Francisco, CA.
11. Y Guo, AF Arellano, *C Roychoudhury*, A Sorooshian, R Kumar, G Pfister (2023). **Harnessing our Air Quality Modeling & Observational Capabilities to Establish Key Factors Influencing Ozone Levels in Arizona**. Poster at 2023 MAC-MAQ Conference, UC Davis, CA.
12. MA Mirrezaei, Y Guo, *C Roychoudhury*, AF Arellano, A Sorooshian, W Tang, L Emmons (2023). **Investigating surface ozone sensitivity to HCHO/NO_2 ratios over Arizona using the Multi-Scale**

Infrastructure for Chemistry and Aerosols (MUSICA) model. Poster at 2023 MAC-MAQ Conference, UC Davis, CA.

13. D Das, S Chiao, ET Swenson, GG Persad, *C Roychoudhury* (2023). **Past, Present and Future Humid Heat Extremes over the East Coast of the United States** (2023). Poster at 2023 103rd AMS Annual Meeting, Denver, CO.
14. *C Roychoudhury*, C He, R Kumar, JM McKinnon and AF Arellano (2022). **Tracing the sources of black carbon deposition over the glaciers in High Mountain Asia: A tagged-tracer approach using WRF-Chem**. Poster at 2022 AGU Fall Meeting, Chicago, IL.
15. JM McKinnon, AF Arellano, *C Roychoudhury* (2022). **Spatio-temporal Pattern Analysis of Trace Gases and Aerosol Abundance Using Varimax Rotation and Locally Linear Embeddings.** Poster at 2022 AGU Fall Meeting, Chicago, IL.
16. *C Roychoudhury*, C He, R Kumar, JM McKinnon and AF Arellano (2022). **Source attribution of aerosol impacts to snow cover over High Mountain Asia.** Poster at 2022 International Global Atmospheric Chemistry (IGAC) Project Science Conference, Manchester, UK.
17. *C Roychoudhury*, C He, R Kumar, MK Shrivastava, JM McKinnon , AF Arellano (2022). **Do aerosols really matter over High Mountain Asia?**. Oral Presentation at University of Arizona's annual El Dia Del Agua Y La Atmósfera, Tucson, AZ.
18. *C Roychoudhury*, C He, R Kumar, and AF Arellano (2021). **Investigating the relationship of meteorology and atmospheric composition to snow cover: A comparative study over High-Mountain Asia and Andes**. Lightning Talk and Poster at 2021 International Global Atmospheric Chemistry (IGAC) Project Science Conference, (virtual).
19. *C Roychoudhury*, C He, R Kumar and AF Arellano (2021). **Exploring the association of meteorology and atmospheric composition to snow cover changes: A case study over High-Mountain Asia and Central Andes**. Lightning Talk at 2021 MAC-MAQ Conference, (virtual).
20. *C Roychoudhury*, C He, R Kumar, MK Shrivastava, JM McKinnon , AF Arellano (2021). **Model simulations and satellite data analysis of aerosol impacts to snow cover over High Mountain Asia**. Oral Talk at 2021 Fall Meeting, AGU, New Orleans, LA.
21. JM McKinnon, *C Roychoudhury*, B Gaubert, RR Buchholz, AF Arellano (2021). **Spatio-temporal Pattern Analysis of Trace Gases and Aerosol Abundance Using PCA, SOMs, and Convolution Autoencoders**. Oral Talk at 2021 AGU Fall Meeting, and 2022 AMS Annual Meeting.
22. C He, R Kumar, MK Shrivastava, *C Roychoudhury*, AF Arellano (2021). **Brown carbon climatic impacts over High Mountain Asia: WRF-Chem model implementation and application**. Presented at 2021 AGU Fall Meeting (virtual).
23. D Das, D Strauss, *C Roychoudhury*, E Swenson, S Paul, G Fang, P Sinha, A Roy Chowdhury (2020). **Oceanic and Atmospheric factors contributing towards the rapid intensification of tropical cyclones in a warming climate: A diagnostic study of Super Cyclone AMPHAN over the Bay of Bengal**. Poster at 2020 AGU Fall Meeting (virtual).
24. D Das, *C Roychoudhury*, S Paul, F Khan, S Chaudhuri (2020). **Impact of ENSO on Tropical Cyclone Season over North Indian Ocean**. Oral Presentation at the International Virtual Conference on Earth's Changing Climate: Past, Present & Future, Society of Earth Scientists (virtual).
25. F Khan, D Das, *C Roychoudhury*, S Chaudhuri (2018). **Role of geo-potential height in estimating the variablity in Indian Summer Monsoon Rainfall: A comparative study with NCEP-NCAR Reanalysis and CFSR**. Poster Presentation at 2018 TROPMET National Symposium, Indian Meteorological Society.
26. *C Roychoudhury*, R Ray (2018). **Impact of climate change on butterfly population over a metropolis of India**. Oral Presentation at 2018 TROPMET National Symposium, Indian Meteorological Society and BIOSPECTRUM-2018, India.

Technical Skills

Programming Python, GrADS, IDL, MATLAB, QGIS/ArcGIS, PostgreSQL, LaTeX, Linux, HPC.
Modeling WRF-Chem, CESM/CESM-SCAM, MPAS.

Honors & Awards

i) Recipient of John & Margaret Harshbarger Scholarship, University of Arizona (2024).
ii) Recipient of Sol Resnick Scholarship, University of Arizona (2023).
iii) Recipient of Galileo Circle Scholarship, University of Arizona (2022).
iv) NSF NCAR Graduate Visitor Program (GVP) (2022).
v) Rank 1 in MSc in Atmospheric Science (2019) and eligible for INSPIRE-Fellowship under DST, Government of India.

Journal Reviewer

i) Geophysical Research Letters (Wiley).
ii) Theoretical and Applied Climatology (Springer).
iii) Environmental Monitoring and Assessment (Springer).
iv) Journal of Applied Meteorology and Climatology (AMS).
v) Nature Communications (Springer).
vi) Environmental Research Communications (IOP).